Lab protocol Day 4 – colony PCRs to genotype edits

Colony PCR (2 hours):

Materials: 16 PCR tubes, 0.02 M NaOH, sterile water, Sapphire Mix, amplicon primers at 10 µM, Qubit tubes + dsDNA BR solution, agarose, 4 sequencing tubes
Equipment: vortex, tabletop centrifuge, Qubit device, electrophoresis equipment

Select 8 colonies per setup and do colony PCR using amplicon primers. Make sure to adjust Tm and extension time.

1. Pick a single colony into 50 µL 0,02 M NaOH.
2. Boil at 96°C for 10 min.
3. Spin down briefly in a mini tabletop centrifuge, then use the supernatant as template for the PCR reaction.
4. Make 1.1*x master mix (x=16)

Sapphire 2X master mix	12.5 µl
FWD primer, 10 µM	1 µl
REV primer, 10 µM	1 µl
sterile water	8.5 µl
Template	2 µl

Colony PCR primers:
FWD: o2518 5' AACAGGCAGCAGAGCGTTTG 3'
REV: o2519 5' GCCATCGGAAAGCGGAATAC 3'

5. Place tubes in thermocycler and run program.

Temperature	Time	Number of cycles
98°C	1 min	Single cycle
98°C	5 s	10 cycles
56.5°C	5 s	
72°C	10 s	
98°C	5 s	20 cycles
58.5°C	5 s	
72°C	10 s	
72°C	1 min	Single cycle

6. Make gel and load samples (same protocol as Course_day2).
7. Check for correct band under UV light (PTC: 1348 bp, KO: 257 bp).
8. Select 4 PTC samples and clean up PCRs using 1X ratio beads (same protocol as Course_day1).
9. Determine concentration using the Qubit device (same protocol as Course_day1).
10. Add +- 50 ng sample with 2.5 µl o2518 (10 µM) and fill up with MiliQ till 10 µl in sequencing tubes.
11. Write down number of sequencing kit + position of samples and fill them in on the Eurofins website (**done by Vonesch lab**).